

ST BENEDICT'S COLLEGE



SUBJECT
GRADE
EXAMINER
NAME
CLASS

AP Mathematics- Algebra
11
Mr Benecke

DATE
MARKS
MODERATOR
DURATION

08 July 2017
130
Mrs Povall and Mrs Sillman
2 hours

QUESTION	DESCRIPTION	MAXIMUM MARK	ACTUAL MARK
1	Equations & Inequalities	34	
2	Complex Equations	18	
3	Absolute Values	30	
4	Functions	18	
5	Split Graphs	25	
6	Word Problem	5	
	TOTAL	130	

INSTRUCTIONS:

1. This paper consists of 6 questions and 15 pages.
2. Read the questions carefully.
3. You may use an approved non-programmable and non-graphical calculator, unless otherwise stated.
4. Round off your answers to two decimal digits where necessary.
5. It is in your interest to write legibly and present your work neatly.

QUESTION 1**34 MARKS**

Solve for x (given that $x \in R$) in the following equations:

a) $\frac{x^2-7}{x^3-1} = \frac{1}{x^2+x+1}$

5

b) $\sqrt{2-x} = \frac{2\sqrt{2-x}}{2-x} + 1$

8

c) Decompose $\frac{3x+11}{x^2-x-6}$ into partial fractions.

8

d) $\frac{3x}{2x+6} \leq 0$

5

e) $\frac{5}{x+1} - \frac{x-1}{x^2+1} \leq 0$

8

QUESTION 2

18 MARKS

For the following questions, $x \in \mathbb{C}$

- a) Given $x(x + 6) = n$, determine the values of n for which the equation has complex roots. 4

- b) If $2 + 3i$ is a root of $x^3 - 7x^2 + 25x - 39 = 0$, find the other two roots. 7

c) Determine all the solutions of $x^3 + x + 10 = 0$ if $x = -2$ is a root.

7

QUESTION 3**30 MARKS**

a) Given $3 - |2x - 3| = 1$, solve for x .

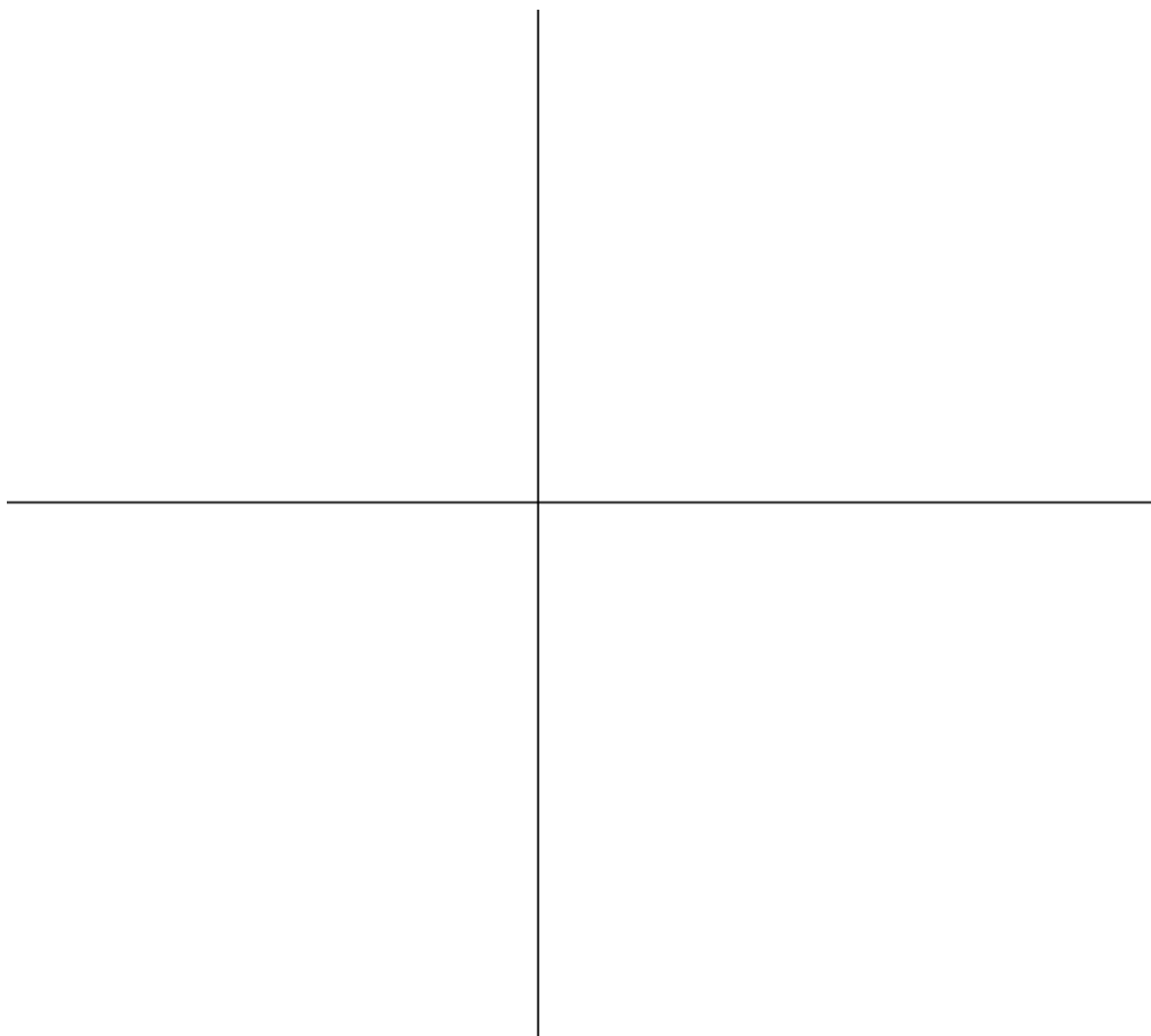
6

b) Given $3 - |2x - 3| = 4$, solve for x .

3

- c) Graph neat sketches of $f(x) = -|x + 2| - 1$ and $g(x) = |-x^2 + 2|$ on the axes provided below.

9



d) Solve the following inequalities

8

(i) $\frac{1}{2}x - 1 > |x + 2| - 3$

(ii) $|x^2| < |x|$

4

QUESTION 4**18 MARKS**

Given functions $f(x) = x^2 + 2$ and $g(x) = 3x + 1$, determine the following and write your answer in simplest form:

a) $f \circ g$

3

b) $g \circ f$

3

c) $f(g(2))$

4

d) $g(2) \circ f$

3

e) $g^{-1}(2)$

5

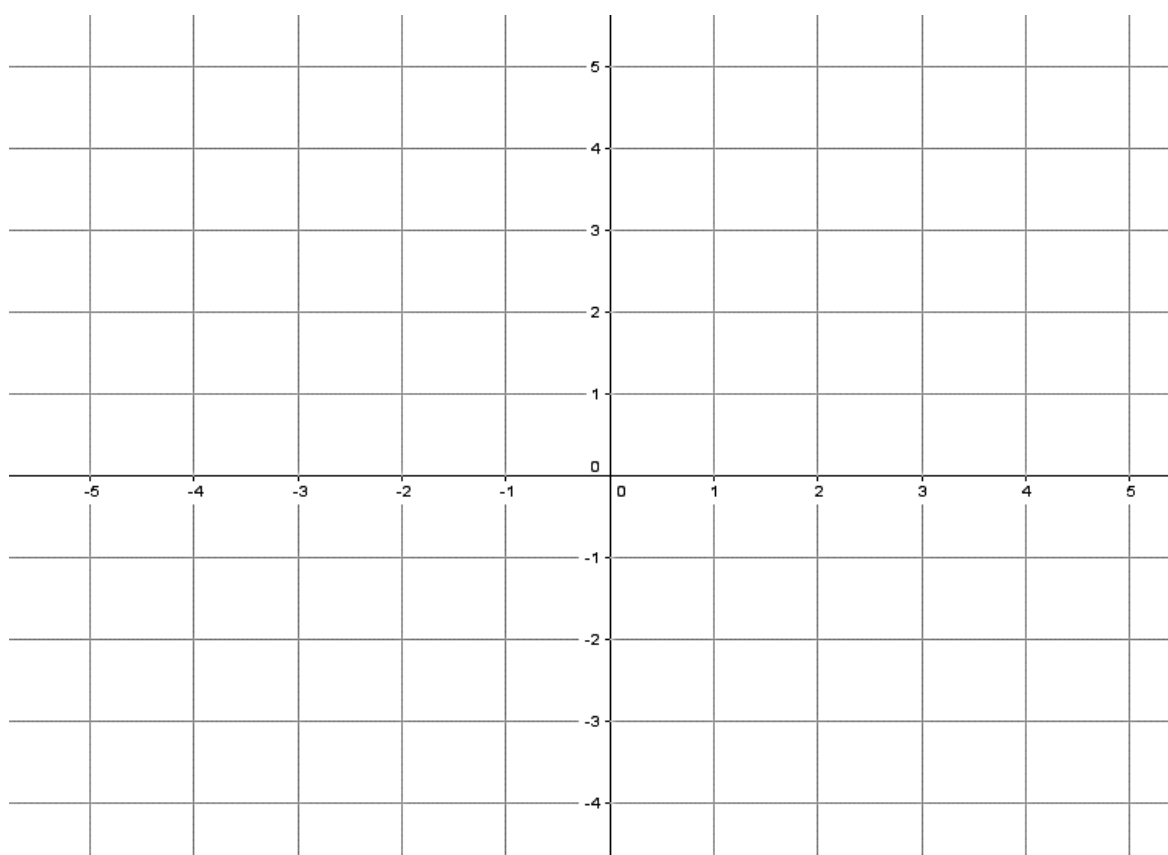
QUESTION 5

25 MARKS

$$\text{Let } f(x) = \begin{cases} |x| & \text{if } x < 0 \\ x^2 & \text{if } 0 < x < 1 \\ 2x + 1 & \text{if } x \geq 1 \end{cases}$$

a) Sketch $f(x)$ on the axes provided below.

6



b) State the domain and range of the function.

3

c) Determine whether or not the function has any zeroes. If it does, list the zeroes. 2

d) Determine $f(0)$, $f(1)$, and $f(2)$

3

e) For which value(s) of x is $f(x)$ discontinuous? Classify these discontinuities.

4

f) For which values of x is $f(x)$ continuous.

3

g) Redefine $f(x)$ so that it is continuous for $x \in R$. Complete the following:

4

$$f(x) = \begin{cases} |x| & \text{if } \underline{\hspace{1cm}} \\ x^2 & \text{if } \underline{\hspace{1cm}} \\ \underline{\hspace{1cm}} & \text{if } \underline{\hspace{1cm}} \end{cases}$$

QUESTION 6**5 MARKS**

The length of a rectangle is 4 meters more than its width. If 2 meters are taken from the length and added to the width and the figure becomes a square with an area of 361 square meters. What are the dimensions of the original figure?

Formula Sheet

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$|x| = \begin{cases} x & \text{if } x \geq 0 \\ -x & \text{if } x < 0 \end{cases}$$

$$z = a + bi$$

$$z^* = a - bi$$